Independent and dependent events

1 Determine whether the events are independent or dependent:

- **a** A teacher has a prize bag filled with different prizes. The students form a line to draw a prize from the bag at random. Once a student has drawn a prize, they take it back to their desk.
- **b** A card is randomly selected from a normal deck of cards without being replaced, and a coin is tossed.
- **c** In a lottery, a numbered ball is selected from a basket. The result is recorded and the ball is set aside. Several more numbers are taken from the basket in this way until the lottery is complete.
- **d** Two cards are randomly selected one after the other from a normal deck of cards without replacement.
- **e** A card is randomly selected from a normal deck of cards, and then returned to the deck. The deck is shuffled and another card is selected.

2 Ursula takes a bus to the station and then immediately gets on a train to work. Is the probability of her missing the train independent or dependent of her missing her bus?

3 A coin is tossed and lands on tails. Will this affect the chances of the coin landing on tails in the next toss?

4 Valerie owns 9 red shirts and 9 blue shirts. She randomly chooses a shirt for the day, and she gets a red shirt. At the end of the day, she puts it into the washing basket.

- **a** Will this affect the chances of choosing a red shirt tomorrow?
- **b** Will this affect the chances of choosing a blue shirt tomorrow?

5 Two dice are rolled. If one die lands on 4, will this affect the chance of the other die landing on 4?

6 A color wheel with a spinner has red, blue and green sectors. If the spinner is spun and lands on green, will this affect the chance of the spinner landing on green next time?

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7 Charlie is going to randomly select a card from a standard deck of cards.
Determine whether the following events would affect the chances of picking a red Queen next time:

- **a** Randomly selecting a red card from the deck and putting it back.
- **b** Randomly selecting a 4 from the deck and putting it back.
- **c** Randomly selecting a spade from the deck and not putting it back.
- **d** Randomly selecting a black card from the deck and not putting it back.

8 In a game of monopoly, two dice are rolled.

- **a** Is the outcome of the second die dependent on the outcome of the first?
- **b** If a '5' is rolled on one of the dice, find the probability of rolling a '5' on the second die.

9 In a lottery, there are 45 balls.

- **a** Find the probability of a particular ball being drawn first.
- **b** A ball is discarded after it has been drawn. If ball number 35 is drawn on the first go, find the probability of ball number 20 being drawn next.
- **c** Is the probability of each successive ball drawn the same as the probability of the first ball drawn?
- **d** Are the draws dependent or independent events?

10 On a roulette table, a ball can land on one of 18 red or 18 black numbers.

- **a** If it lands on a red number on the first go, what is the probability it will land on a red number on the second go?
- **b** Are the successive events of twice landing on a red ball dependent or independent?
- **c** In a game of roulette, the ball has landed on a black number 6 times in a row. Is it highly likely, equally likely or highly unlikely that the ball will land on black the next go?

11 A standard deck of cards is shuffled and placed face down. Dylan attempts to draw five cards randomly so as to get a royal flush (10, Jack, Queen, King, Ace all of the same suit) .

- **a** If he draws a Queen of spades on the first go, find the probability that he draws a 10 of spades on the second go.
- **b** If he draws a Queen of spades on the first go and a 10 of spades on the second go, find the probability that he draws a Jack of spades on the third go.
- **c** Is the probability of each draw dependent or independent of the previous draw?

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12 Vanessa has 55 songs in a playlist. She starts to play them on shuffle so each song will be played once until all songs have been played.

a Find the probability of Song A coming on first.

b Find the probability of Song A coming on last.

c Is the probability of a particular song playing dependent or independent of the position it plays in?

13 In a memory game, 16 pairs of identical cards are randomly placed face down. When someone has a go, they turn one card over and then turn a second card over to try to find an identical pair. If an identical pair are found, they are removed from the pack.

a At the beginning of the game, Kathleen turns a card over. Find the probability that she will pick a card that will make an identical pair.

b The first two cards Kathleen picks are identical. Is the probability of picking any more pairs of identical cards independent or dependent on the number of pairs picked before?

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